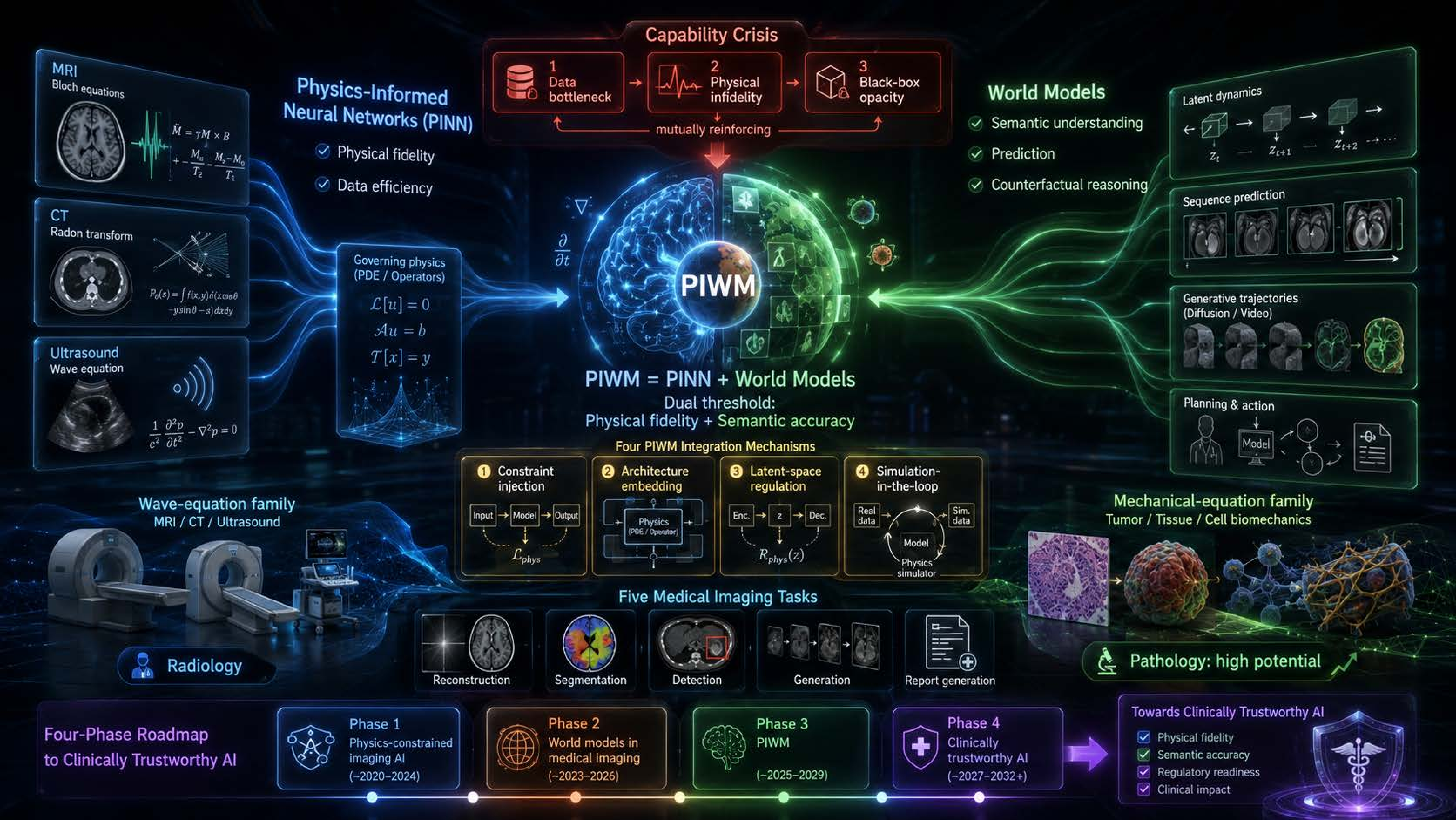
Capability Crisis
1 Data bottleneck
2 Physical infidelity
3 Black-box opacity
mutually reinforcing
MRI
Bloch equations
CT
Radon transform
Ultrasound
Wave equation
Physics-Informed Neural Networks (PINN)
Physical fidelity
Data efficiency
Governing physics (PDE / Operators)
$\mathcal{L}[u] = 0$
$\mathcal{A}u = b$
$\mathcal{T}[x] = y$
PIWM
World Models
Semantic understanding
Prediction
Counterfactual reasoning
Latent dynamics
Sequence prediction
Generative trajectories (Diffusion / Video)
Planning & action
Model
PIWM = PINN + World Models
Dual threshold:
Physical fidelity + Semantic accuracy
Four PIWM Integration Mechanisms
1 Constraint injection
Input
Model
Output
$\mathcal{L}_{phys}$
2 Architecture embedding
Physics (PDE / Operator)
3 Latent-space regulation
Enc.
z
Dec.
$R_{phys}(z)$
4 Simulation-in-the-loop
Real data
Sim. data
Model
Physics simulator
Wave-equation family
MRI / CT / Ultrasound
Mechanical-equation family
Tumor / Tissue / Cell biomechanics
Five Medical Imaging Tasks
Reconstruction
Segmentation
Detection
Generation
Report generation
Radiology
Pathology: high potential
Four-Phase Roadmap to Clinically Trustworthy AI
Phase 1
Physics-constrained imaging AI
(~2020–2024)
Phase 2
World models in medical imaging
(~2023–2026)
Phase 3
PIWM
(~2025–2029)
Phase 4
Clinically trustworthy AI
(~2027–2032+)
Towards Clinically Trustworthy AI
Physical fidelity
Semantic accuracy
Regulatory readiness
Clinical impact